

EXP:11

```

1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.tree import DecisionTreeClassifier
4 from sklearn.metrics import accuracy_score
5 X,y=load_iris(return_X_y=True)
6 X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
7 model=DecisionTreeClassifier().fit(X_train,y_train)
8 print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))

```

• Accuracy: 1.0

EXP:12

```

1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestClassifier
4 from sklearn.metrics import accuracy_score
5 X,y=load_iris(return_X_y=True)
6 X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
7 model=RandomForestClassifier().fit(X_train,y_train)
8 print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))

```

• Accuracy: 1.0

EXP:13

```

1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.svm import SVC
4 from sklearn.metrics import accuracy_score
5 X,y=load_iris(return_X_y=True)
6 X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
7 model=SVC().fit(X_train,y_train)
8 print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))

```

*** Accuracy: 1.0

EXP:14

```

1 import numpy as np
2 x=np.array([1,2,3,4,5]); y=np.array([2,4,6,8,10]); m,b,lr=0,0,0.01
3 for i in range(1000): y_pred=m*x+b; dm=(-2/len(x))*sum(x*(y-y_pred)); db=(-2/len(x))*sum(y-y_pred); m-=
4 print("Slope:",m)
5 print("Intercept:",b)

```

*** Slope: 1.9951803506719779
Intercept: 0.017400463340610635

EXP:15

```

import cv2, numpy as np
from google.colab import files; from matplotlib import pyplot as plt
img = cv2.cvtColor(cv2.imread(list(files.upload().keys())[0]), cv2.COLOR_BGR2RGB)
Z = np.float32(img.reshape((-1,3))); k = int(input("Enter K: "))
_, label, center = cv2.kmeans(Z, k, None, (cv2.TERM_CRITERIA_EPS+cv2.TERM_CRITERIA_MAX_ITER,100,0.2), 10, cv2.KMEANS_RANDOM_CENTERS)
result = np.uint8(center[label.flatten()]).reshape(img.shape)
plt.subplot(121); plt.imshow(img); plt.title("Original"); plt.axis("off")
plt.subplot(122); plt.imshow(result); plt.title("Segmented"); plt.axis("off"); plt.show()

```

*** Choose Files WIN_20260..._49_Pro.jpg
 WIN_20260709_10_22_49_Pro.jpg(image/jpeg) - 138602 bytes, last modified: 7/9/2026 - 100% done
 Saving WIN_20260709_10_22_49_Pro.jpg to WIN_20260709_10_22_49_Pro.jpg
 Enter K: 3

Original



Segmented

